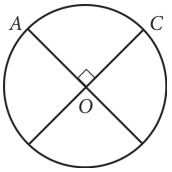
Circles

Geometry and Trigonometry · Math · 67 questions

1 08358-DC · Easy



The circle above with center O has a circumference of 36. What is the length of minor arc $\overset{\frown}{AC}$?

A. 9

B. 12

C. 18

D. 36

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2 0b5a2082-3d77-4acf-86c6-0d878c0c0b93 · Easy

A circle has a diameter of 12.6 centimeters. What is the radius of the circle, in centimeters?

Student-produced response: _____

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3 5235bd99-6451-4d06-9e8e-2810669a0c43 · Easy

$$x - 6^2 + y - 3^2 = 81$$

The graph of the given equation in the xy-plane is a circle. What is the length of the radius of this circle?

A. 3

B. 6

C. 9

D. 81

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4 111ca09a-c599-4fa2-82b2-aee6b59db1ea · Easy

What is the value of $\cos\frac{565\pi}{6}$?

A. $\frac{1}{2}$

B. 1

C. $\frac{\sqrt{3}}{2}$

D. $\sqrt{3}$

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5 8440c7b5-f1af-40ac-839d-ad1fdfe3d955 · Medium

What is the value of $\sin 42\pi$?

A. 0

B. $\frac{1}{2}$

C. $\frac{\sqrt{2}}{2}$

D. 1

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6 94dce764-2378-4b26-b4b0-980f97c7224f · Medium

The measure of angle Z is $60°$. What is the measure, in radians, of angle Z?

A. $\frac{1}{6}\pi$

B. $\frac{1}{3}\pi$

C. $\frac{2}{3}\pi$

D. 1π

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7 00581-DC · Medium

The number of radians in a 720-degree angle can be written as $a\pi$, where a is a constant. What is the value of a ?

Student-produced response: _____

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8 28906692-49f1-4c04-95f4-a0cc239c8408 · Medium

What is the radius of the circle in the xy-plane defined by $x + 3^2 + y + 5^2 = 289$?

Student-produced response: _____

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9 408558ac-8ac7-4c5e-ad29-a40f145bdcfb · Medium

An angle has a measure of $\frac{9\pi}{20}$ radians. What is the measure of the angle in degrees?

Student-produced response: _____

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10 10d4dc39-2d6d-4fb6-8cc4-828292d3cbe0 · Medium

A circle in the *xy*-plane has its center at $-4, 5$ and the point $-8, 8$ lies on the circle. Which equation represents this circle?

A. $x - 4^2 + y + 5^2 = 5$

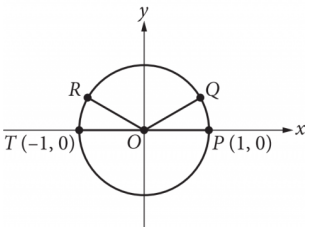
B. $x + 4^2 + y - 5^2 = 5$

C. $x - 4^2 + y + 5^2 = 25$

D. $x + 4^2 + y - 5^2 = 25$

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11 025648-DC · Medium



In the xy-plane above, points P, Q, R, and T lie on the circle with center O. The degree measures of angles POQ and ROT are each 30°. What is the radian measure of angle QOR ?

A. $\frac{5}{6}\pi$

B. $\frac{3}{4}\pi$

C. $\frac{2}{3}\pi$

D. $\frac{1}{3}\pi$

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12 47a97bc9-80a5-4ac1-9410-742cea3e5ba7 · Medium

A circle in the xy-plane has the equation $(x - 19)^2 + (y - 17)^2 = 64k^2$, where k is a positive constant. Which of the following gives the center of the circle and its radius?

A. The center is at $(19, 17)$, and the radius is 8.

B. The center is at $(19, 17)$, and the radius is $8k$.

C. The center is at $(19, 17)$, and the radius is 64.

D. The center is at $(19, 17)$, and the radius is $64k$.

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13 aa85b05d-ab61-4169-aad8-c85d16b1e1c9 · Medium

$$x - 3^2 + y + 16^2 = 289$$

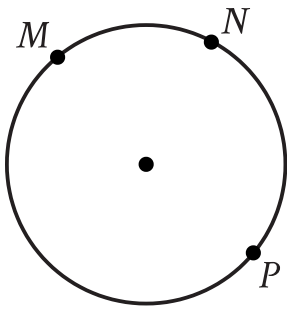
The graph of the given equation is a circle in the xy-plane. The point q, s lies on the circle. Which of the following is NOT a possible value of q?

A. -17

B. -3

C. 3

D. 17



- Clockwise from top left, the following points are on the circle:
  - upper M
  - upper N
  - upper P

Points M, N, and P lie on the circle shown. On this circle, minor arc MN has a length of 39 centimeters and major arc MPN has a length of 195 centimeters. What is the circumference, in centimeters, of the circle shown?

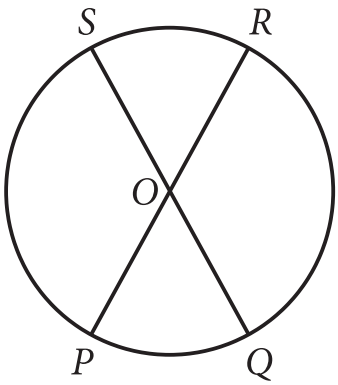
A. 39

B. 156

C. 195

D. 234

15 28b7a76a-9194-4446-b1c9-e00fb73bc3c8 · Medium



Note: Figure not drawn to scale.

- The center of the circle is point upper O.
- Points upper S, upper R, upper Q, and upper P are on the circle.
- Line segment upper P upper R is a diameter of the circle.
- Line segment upper Q upper S is a diameter of the circle.
- Diameters upper P upper R and upper Q upper S intersect at point upper O.
- A note indicates the figure is not drawn to scale.

The circle shown has center O, circumference 144π, and diameters $\bar{PR}$ and $\bar{QS}$. The length of arc PS is twice the length of arc PQ. What is the length of arc QR?

A. 24π

B. 48π

C. 72π

D. 96π

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16 0d7ac734-4775-44b6-bcf4-405ac5ab2009 · Medium

What is the radius of the circle in the xy-plane defined by $x + 2^2 + y + 5^2 = 169$?

Student-produced response: _____

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17 ab88ac2a-4efa-4ff6-86c0-5005fa88a598 · Medium

The difference between the measure of angle A and the measure of angle B is $-\frac{5}{6}\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B, in degrees?

A. $-\frac{5}{6}\pi \cdot \frac{360°}{\pi}$

B. $-\frac{5}{6}\pi \cdot \frac{180°}{\pi}$

C. $-\frac{5}{6}\pi \cdot \frac{\pi}{180°}$

D. $-\frac{5}{6} \cdot \frac{180°}{\pi}$

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18 028868-DC · Medium

In the xy-plane, a circle with radius 5 has center $(-8,6)$. Which of the following is an equation of the circle?

A. $(x-8)^2+(y+6)^2=25$

B. $(x+8)^2+(y-6)^2=25$

C. $(x-8)^2+(y+6)^2=5$

D. $(x+8)^2+(y-6)^2=5$

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19 889ad56f-33a9-4c3b-b4d8-24aba4870171 · Medium

An angle has a measure of $\frac{16\pi}{15}$ radians. What is the measure of the angle, in degrees?

Student-produced response: _____

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20 c2d6ad1f-5e60-4cc7-a1e8-cc677271772b · Medium

A circle in the *xy*-plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

A. The center is at $(13, k)$ and the radius is 8.

B. The center is at $(k, 13)$ and the radius is 8.

C. The center is at $(k, 13)$ and the radius is 64.

D. The center is at $(13, k)$ and the radius is 64.

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21 d79e2a0a-e6b4-473c-86f6-8d246a9ae3e9 · Medium

In the *xy*-plane, the graph of the equation $x - 3^2 + y - 5^2 = 9$ is a circle. The point $6, c$, where c is a constant, lies on this circle. What is the value of c?

Student-produced response: _____

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22 735dd150-97a4-4853-addc-1b4ab44812db · Medium

$$x + 4^2 + y - 19^2 = 121$$

The graph of the given equation is a circle in the *xy*-plane. The point a, b lies on the circle. Which of the following is a possible value for a?

A. -16

B. -14

C. 11

D. 19

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23 6899531e-bbaf-4fee-900a-53c4fc396050 · Medium

$$x^2 + 58x + y^2 = 0$$

In the xy-plane, the graph of the given equation is a circle. What are the coordinates x, y of the center of the circle?

A. $0, 29$

B. $0, -29$

C. $29, 0$

D. $-29, 0$

---

24 0f1a7b12-733e-45ba-a7fc-021a951885f5 · Medium

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R. What is the measure of angle T, in degrees?

A. 75

B. 120

C. 195

D. 390

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25 87f88638-07ba-4370-ae6b-d743f0fba4ad · Medium

Circle A has equation $x - 7^2 + y + 3^2 = 1$. In the xy-plane, circle B is obtained by translating circle A to the right 4 units. Which equation represents circle B?

A. $x - 7^2 + y + 7^2 = 1$

B. $x - 3^2 + y + 3^2 = 1$

C. $x - 11^2 + y + 3^2 = 1$

D. $x - 7^2 + y - 1^2 = 1$

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26 af30d73b-01f0-4546-bd51-0d5af9b64d57 · Medium

What is the center of the circle in the *xy*-plane defined by the equation $x - 1^2 + y + 7^2 = 1$?

A. $-1, -7$

B. $-1, 7$

C. $1, -7$

D. $1, 7$

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27 d5ce6f43-c646-4b30-bf82-ba8eaf2f70bf · Hard

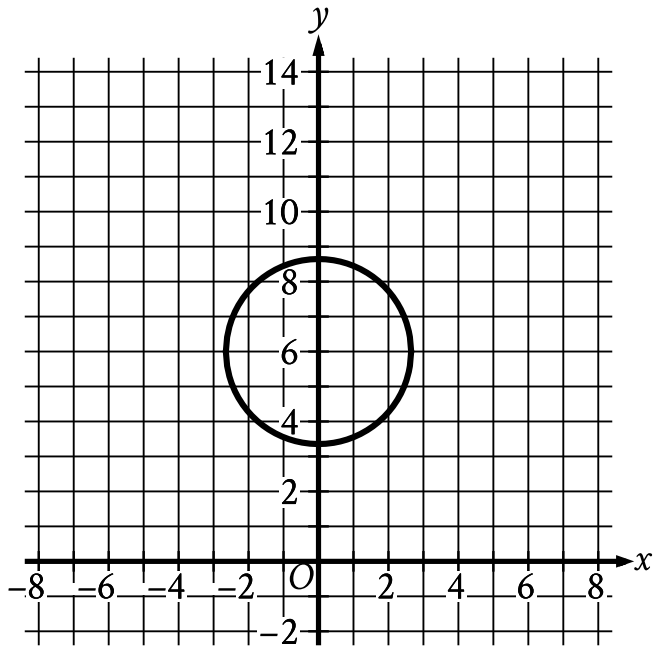
A circle has center *G*, and points *M* and *N* lie on the circle. Line segments *MH* and *NH* are tangent to the circle at points *M* and *N*, respectively. If the radius of the circle is 168 millimeters and the perimeter of quadrilateral *GMHN* is $3{,}856$ millimeters, what is the distance, in millimeters, between points *G* and *H*?

A. 168

B. $1{,}752$

C. $1{,}760$

D. $1{,}768$



- The circle passes through the following approximate points:
  - (0 comma 8.6)
  - (2.6 comma 6)
  - (0 comma 3.4)
  - (negative 2.6 comma 6)

Circle A shown is defined by the equation $x^2 + y - 6^2 = 7$. Circle B (not shown) has the same radius but is translated 96 units to the right. If the equation of circle B is $x - h^2 + y - k^2 = a$, where h, k, and a are constants, what is the value of $4a$?

Student-produced response: _____

29 bf1e177f-7c4f-4d1a-9d5d-36f0191c1cde · Hard

A circle in the *xy*-plane has its center at $-4, -6$. Line *k* is tangent to this circle at the point $-7, -7$. What is the slope of line *k*?

A. -3

B. $-\frac{1}{3}$

C. $\frac{1}{3}$

D. 3

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30 9497dfe2-0a1a-4668-81ab-c2c1ec879a50 · Hard

Circle A in the *xy*-plane has the equation $x + 5^2 + y - 5^2 = 25$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the *xy*-plane is $x + 5^2 + y - 5^2 = k$, where *k* is a constant. What is the value of *k*?

Student-produced response: _____

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31 a786b147-72b1-4d1e-b373-354a8186f3a8 · Hard

The equation $x^2 + y - 1^2 = 49$ represents circle A. Circle B is obtained by shifting circle A down 2 units in the *xy*-plane. Which of the following equations represents circle B?

A. $x - 2^2 + y - 1^2 = 49$

B. $x^2 + y - 3^2 = 49$

C. $x + 2^2 + y - 1^2 = 49$

D. $x^2 + y + 1^2 = 49$

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32 22d64b0b-de62-4d03-bfd1-a0f9bfc7fb43 · Hard

Points Q and R lie on a circle with center P. The radius of this circle is 9 inches. Triangle PQR has a perimeter of 31 inches. What is the length, in inches, of $\bar{QR}$?

A. $13\sqrt{2}$

B. 13

C. $9\sqrt{2}$

D. 9

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33 78306704-22d8-4501-a060-e7b7995c3ceb · Hard

Which of the following equations represents a circle in the xy-plane that intersects the y-axis at exactly one point?

A. $(x-8)^2+(y-8)^2=16$

B. $(x-8)^2+(y-4)^2=16$

C. $(x-4)^2+(y-9)^2=16$

D. $x^2+(y-9)^2=16$

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34 01735-DC · Hard

In the xy-plane, the graph of $2x^2-6x+2y^2+2y=45$ is a circle. What is the radius of the circle?

A. 5

B. 6.5

C. $\sqrt{40}$

D. $\sqrt{50}$

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35 7ceef3d0-2aeb-40e2-a3fe-45da4ef0f952 · Hard

The measure of angle *L* is 60 degrees. If the measure of angle *L* is $\frac{\pi}{n}$ radians, where *n* is a constant, what is the value of *n*?

Student-produced response: _____

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36 69d54aad-89b3-418d-8c94-c011d9fe97fd · Hard

The measure of angle *A* is $\frac{\pi}{216}$ radians. The measure of angle *B* is 36 times the measure of angle *A*. What is the value of $\cos B$?

A. 0

B. $\frac{1}{2}$

C. $\frac{\sqrt{2}}{2}$

D. $\frac{\sqrt{3}}{2}$

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37 437a1594-67bd-4051-8026-2a747bd29288 · Hard

A circle has center *O*, and points *R* and *S* lie on the circle. In triangle *ORS*, the measure of $\angle ROS$ is $88°$. What is the measure of $\angle RSO$, in degrees? (Disregard the degree symbol when entering your answer.)

Student-produced response: _____

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38 06441198-c645-4566-b641-e692fa31e1ff · Hard

In the *xy*-plane, circle *F* is defined by the equation $x + 8^2 + y + 8^2 = 25$. Circle *G* has the same center as circle *F*, and the radius of circle *G* is 2 units greater than the radius of circle *F*. Circle *G* is defined by the equation $x + 8^2 + y + 8^2 = p$, where *p* is a constant. What is the value of *p*?

Student-produced response: _____

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39 1c2c5aa2-c601-4e42-9804-f7298eb7c510 · Hard

Point *F* lies on a unit circle in the *xy*-plane and has coordinates $1, 0$. Point *G* is the center of the circle and has coordinates $0, 0$. Point *H* also lies on the circle and has coordinates -1, *y*, where *y* is a constant. Which of the following could be the positive measure of angle *FGH*, in radians?

A. $\frac{27\pi}{2}$

B. $\frac{29\pi}{2}$

C. 24π

D. 25π

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40 62601029-202e-4008-8a43-a7e84f34406d · Hard

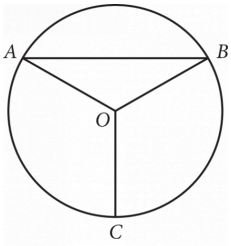
$$x^2 + y^2 - 10x - 8y - 14n = 0$$

The given equation represents circle A in the *xy*-plane, where *n* is a constant. Point 118 lies on circle B, which has the same center but twice the diameter as circle A. What is the value of *n*?

Student-produced response: _____

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41 11305-DC · Hard



Point O is the center of the circle above, and the measure of $\angle OAB$ is $30°$. If the length of $\overline{OC}$ is 18, what is the length of arc $\overset{\frown}{AB}$?

A. 9π

B. 12π

C. 15π

D. 18π

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42 301edaf1-d774-4eda-96b8-80c08070887f · Hard

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy-plane is a circle. What is the length of the circle's radius?

Student-produced response: _____

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43 f85fd217-961b-4d79-9d31-21aca15bd9e2 · Hard

In the xy-plane, a circle has center C with coordinates h, k. Points A and B lie on the circle. Point A has coordinates $h + 1, k + \sqrt{102}$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

A. $\sqrt{206}$

B. $2\sqrt{102}$

C. $103\sqrt{2}$

D. $103\sqrt{3}$

---

44 97ba1ce7-b7c6-4164-bd91-cbae30d8e22a · Hard

What is the value of $\tan\frac{92\pi}{3}$?

A. $-\sqrt{3}$

B. $-\frac{\sqrt{3}}{3}$

C. $\frac{\sqrt{3}}{3}$

D. $\sqrt{3}$

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45 84bcf173-28fe-47bf-bf15-075592753ee0 · Hard

Point *O* is the center of a circle. The measure of arc *RS* on this circle is $100°$. What is the measure, in degrees, of its associated angle *ROS*?

Student-produced response: _____

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46 e7973fae-0e6b-4fe9-bac8-a13801f05c72 · Hard

A circle in the *xy*-plane has its center at $-1, 1$. Line *t* is tangent to this circle at the point $5, -4$. Which of the following points also lies on line *t*?

A. $0, \frac{6}{5}$

B. $4, 7$

C. $10, 2$

D. $11, 1$

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47 93083ace-02bf-4d1e-9794-d51ec7697e04 · Hard

A circle in the xy-plane has its center at $(16, 17)$ and has a radius of $7k$. Which equation represents this circle?

A. $(x - 16)^2 + (y - 17)^2 = 49k$

B. $(x - 16)^2 + (y - 17)^2 = 49k^2$

C. $(x - 16)^2 + (y - 17)^2 = 7k$

D. $(x - 16)^2 + (y - 17)^2 = 7k^2$

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48 cf2ac41c-6770-4663-bc5c-fdcc886db696 · Hard

$$x^2 + 14x + y^2 = 6y + 109$$

In the xy-plane, the graph of the given equation is a circle. What is the length of the circle's radius?

A. $\sqrt{109}$

B. $\sqrt{149}$

C. $\sqrt{167}$

D. $\sqrt{341}$

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49 0727e22f-f583-4696-b931-a7ab10b22425 · Hard

$$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$$

The given equation, where t is a constant, defines a circle in the xy-plane. The radius of this circle is $\sqrt{83}$. What is the value of t?

A. 73

B. 71

C. 37

D. 35

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50 c744c3b6-b2fb-433d-8650-de2dcf230cc9 · Hard

A circle has center P, and points A and B lie on the circle. The measure of arc AB is $45°$ and the length of arc AB is 4π units. What is the length, in units, of the radius of the circle?

Student-produced response: _____

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51 11301-DC · Hard

$(x-6)^2+(y+5)^2=16$

In the xy-plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates $(10,-5)$. If $\overline{PQ}$ is a diameter of the circle, what are the coordinates of point Q ?

A. $(2,-5)$

B. $(6,-1)$

C. $(6,-5)$

D. $(6,-9)$

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52 08283-DC · Hard

$x^2+20x+y^2+16y=-20$

The equation above defines a circle in the xy-plane. What are the coordinates of the center of the circle?

A. $(-20,-16)$

B. $(-10,-8)$

C. $(10,8)$

D. $(20,16)$

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53 51707cf8-d699-4a87-a4c7-1cf41c8e4d28 · Hard

A circle in the *xy*-plane has its center at $-5, 2$ and has a radius of 9. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where *a*, *b*, and *c* are constants. What is the value of *c*?

Student-produced response: _____

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54 96b9ead2-bb18-42ec-ac49-2021473157bc · Hard

In the *xy*-plane, circle *M* is the graph of the equation $x - 4^2 + y - 5^2 = 9$. Circle *P* has the same center as circle *M* but has a radius that is twice the radius of circle *M*. Which equation represents circle *P*?

A. $x - 4^2 + y - 5^2 = 18$

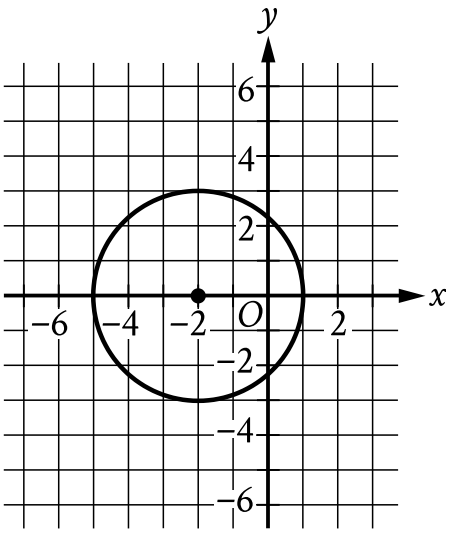
B. $x - 4^2 + y - 5^2 = 36$

C. $x - 8^2 + y - 10^2 = 9$

D. $x - 8^2 + y - 10^2 = 18$

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55 a17099a6-ebb4-4a44-aaf9-5dd8051a2c25 · Hard



- The center of the circle is the point (negative 2 comma 0).
- Clockwise from top, the circle passes through the following points:
  - (negative 2 comma 3)
  - (1 comma 0)
  - (negative 2 comma negative 3)
  - (negative 5 comma 0)

Circle A (shown) is defined by the equation $x + 2^2 + y^2 = 9$. Circle B (not shown) is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A. Which equation defines circle B?

A. $x + 2^2 + y + 6^2 = 49$

B. $2x + 2^2 + 2y + 6^2 = 9$

C. $x + 2^2 + y - 6^2 = 49$

D. $2x + 2^2 + 2y - 6^2 = 9$

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56 025406-DC · Hard

The equation $(x+6)^2+(y+3)^2=121$ defines a circle in the xy-plane. What is the radius of the circle?

Student-produced response: _____

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57 10166-DC · Hard

A circle in the xy-plane has equation $(x+3)^2+(y-1)^2=25$. Which of the following points does NOT lie in the interior of the circle?

A. $(-7, 3)$

B. $(-3, 1)$

C. $(0, 0)$

D. $(3, 2)$

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58 20c5df6f-78ac-41d4-99b0-3fda967fa4b0 · Hard

What is the diameter of the circle in the *xy*-plane with equation $x - 5^2 + y - 3^2 = 16$?

A. 4

B. 8

C. 16

D. 32

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59 a8b375f5-371f-458f-bca7-e5b78b360f11 · Hard

Circle A in the *xy*-plane has the equation $x + 5^2 + y - 5^2 = 4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the *xy*-plane is $x + 5^2 + y - 5^2 = k$, where k is a constant. What is the value of k?

Student-produced response: _____

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60 19517d3f-5560-4607-a8d2-e73bd6c13c56 · Hard

A circle has center O, and points A and U lie on the circle. Line segments AE and UE are tangent to the circle at points A and U, respectively. The distance between point O and point E is 510 centimeters and the distance between point O and point U is 78 centimeters. What is the area, in square centimeters, of quadrilateral $OAEU$?

Student-produced response: _____

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61 a8c9a7ac-0d28-45aa-b013-1691f98acb24 · Hard

A circle in the xy-plane has its center at $(19, 18)$ and has a radius of $9k$. Which equation represents this circle?

A. $(x - 19)^2 + (y - 18)^2 = 81k$

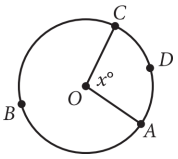
B. $(x - 19)^2 + (y - 18)^2 = 81k^2$

C. $(x - 19)^2 + (y - 18)^2 = 9k$

D. $(x - 19)^2 + (y - 18)^2 = 9k^2$

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62 08415-DC · Hard



The circle above has center O, the length of arc $\overset{\frown}{ADC}$ is 5π, and $x = 100$. What is the length of arc $\overset{\frown}{ABC}$?

A. 9π

B. 13π

C. 18π

D. $\frac{13}{2}\pi$

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63 1597b05c-5fdb-4f34-a5b2-e92fa92dc557 · Hard

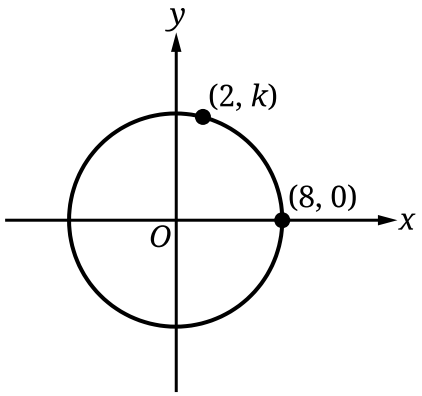
A circle has center O, and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 3 inches. What is the circumference, in inches, of the circle?

A. 3

B. 6

C. 9

D. 24



Note: Figure not drawn to scale.

- The center of the circle is the origin.
- The following points are on the circle:
  - (2 comma k)
  - (8 comma 0)
- A note indicates the figure is not drawn to scale.

The circle shown has its center at $(0, 0)$. What is the value of k?

A. 6

B. 7

C. 8

D. $\sqrt{60}$

65 29baf634-1e61-4b33-9cbd-19d88be26e72 · Hard

The equation $x^2 + y - 2^2 = 36$ represents circle A. Circle B is obtained by shifting circle A down 4 units in the *xy*-plane. Which of the following equations represents circle B?

A. $x^2 + y + 2^2 = 36$

B. $x^2 + y - 6^2 = 36$

C. $x - 4^2 + y - 2^2 = 36$

D. $x + 4^2 + y - 2^2 = 36$

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66 58f8a50f-ca96-4d35-9e5d-bb5694488742 · Hard

A circle in the *xy*-plane has a diameter with endpoints $2, 4$ and $2, 14$. An equation of this circle is $x - 2^2 + y - 9^2 = r^2$, where *r* is a positive constant. What is the value of *r*?

Student-produced response: _____

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67 070632-DC · Hard

Points A and B lie on a circle with radius 1, and arc $\overset{\frown}{AB}$ has length $\frac{\pi}{3}$. What fraction of the circumference of the circle is the length of arc $\overset{\frown}{AB}$?

Student-produced response: _____
